

ASSIGNMENT

Data Extraction in ETL

Question 1 : Describe different types of data sources used in ETL with suitable examples. ?

ANS- ETL uses different types of data sources to extract information for processing:

1.Flat Files – Simple files like CSV, Excel, or text files.

Example: Employee data stored in `employees.csv`.

2.Relational Databases – Structured tables with relationships.

Example: MySQL or Oracle databases used in business systems.

3.NoSQL Databases – Handle unstructured data such as JSON.

Example: MongoDB for storing product details.

4.APIs – Provide real-time data from web services.

Example: Weather API or payment gateway API.

Question 2 : What is data extraction? Explain its role in the ETL pipeline. ?

ANS- Data extraction is the first step of the ETL (Extract, Transform, Load) process. It involves collecting data from various sources such as databases, flat files, APIs, or web logs. Its role is to pull raw data from these sources and make it ready for transformation—cleaning, filtering, and formatting before loading it into a data warehouse or analytics system.

Example: Extracting customer records from a MySQL database and sales data from an Excel file to combine them for analysis.

Question 3 : Explain the difference between CSV and Excel in terms of extraction and ETL usage. ?

ANS - CSV files store plain text data separated by commas, making them easy to extract and process in ETL tools.

Excel files contain formatted data, formulas, and multiple sheets, so extraction needs special connectors or conversion to CSV.

Question 4 : Explain the steps involved in extracting data from a relational database.

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Step	Explanation
1 Connect to Database	Link your tool or program to the database using login details (username, password, database name).
2 Write SQL Query	Use SQL commands like SELECT to choose which data you want.
3 Run the Query	Execute the command so the database can find and return the data.
4 Fetch Data	Collect the results into your tool or ETL process.
5 Check Data	Make sure the data is correct — no missing or wrong values.
6 Store in Staging Area	Save the raw data temporarily before cleaning or transforming it.
7 Log the Process	Record details like how many rows were extracted and any errors.

Question 5 : Explain three common challenges faced during data extraction.

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Challenge	Simple Explanation	Example
1 Data Quality Issues	Extracted data may have errors, missing values, or duplicates.	Customer emails with typos or blank fields.
2 Inconsistent Formats	Data comes from different sources with different formats, making it hard to combine.	Dates stored as DD/MM/YYYY in one file and MM-DD-YYYY in another.
3 Large Volume of Data	Handling huge datasets can be slow and resource-heavy.	Extracting millions of sales records from a database.

Question 6 : What are APIs? Explain how APIs help in real-time data extraction

ANS- API means Application Programming Interface. It helps two software systems talk to each other and share data.

Point	Explanation
1 Live Data Access	APIs give data instantly from the source — no need to wait for files.
2 Continuous Updates	They send new data as soon as it changes (like stock prices or sensor readings).
3 Easy to Use Format	APIs share data in simple formats like JSON or XML, which tools can read easily.

Question 7 : Why are databases preferred for enterprise-level data extraction?

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Reason	Explanation	Example
1 Handle Big Data	Databases can manage huge amounts of information without slowing down.	Millions of customer records stored safely.
2 Security & Control	They provide user access control and protect sensitive data.	Only managers can see salary details.
3 Reliability & Consistency	Databases ensure data is accurate and consistent across the company.	Same product price shown in sales, billing, and inventory systems.

Question 8 : What steps should an ETL developer take when extracting data from large CSV files (1GB+)

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Step	Explanation
1 Use Chunking	Don't load the whole file at once. Read it in small parts (chunks) to avoid memory issues.
2 Optimize Tools	Use efficient libraries (like Pandas with chunksize or ETL tools that support streaming).
3 Validate Data Early	Check each chunk for missing values, wrong formats, or duplicates before moving forward.
4 Store in Staging Area	Save processed chunks into a temporary database or staging table for easier handling.
5 Monitor Performance	Keep track of memory usage, speed, and errors during extraction.
6 Use Compression if Needed	Work with compressed formats (like .zip or .gz) to reduce file size and speed up reading.